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REVIEW ARTICLE

Antibiotics, antibiotic-resistant bacteria, and resistance genes in aquaculture: risks, current concern, and future thinking

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Abstract
Aquaculture is remarkably one of the most promising industries among the food-producing industries in the world. Aquaculture production as well as fish consumption per capita have been dramatically increasing over the past two decades. Shifting of culture method from semi-intensive to intensive technique and applying of antibiotics to control the disease outbreak are the major factors for the increasing trend of aquaculture production. Antibiotics are usually present at subtherapeutic levels in the aquaculture environment, which increases the selective pressure to the resistant bacteria and stimulates resistant gene transfer in the aquatic environment. It is now widely documented that antibiotic resistance genes and resistant bacteria are transported from the aquatic environment to the terrestrial environment and may pose adverse effects on human and animal health. However, data related to antibiotic usage and bacterial resistance in aquaculture is very limited or even absent in major aquaculture-producing countries. In particular, residual levels of antibiotics in fish and shellfish are not well documented. Recently, some of the countries have already decided the maximum residue levels (MRLs) of antibiotics in fish muscle or skin; however, many antibiotics are yet not to be decided. Therefore, an urgent universal effort needs to be taken to monitor antibiotic concentration and resistant bacteria particularly multiple antibiotic-resistant bacteria and to assess the associated risks in aquaculture. Finally, we suggest to take an initiative to make a uniform antibiotic registration process, to establish the MRLs for fish/shrimp and to ensure the use of only aquaculture antibiotics in fish and shellfish farming globally.

Keywords Aquaculture · Antibiotics · Resistance risk · Human health risk · Ecological risk · Regulations

Introduction

Aquaculture is one of the fastest growing sectors among the animal production sectors in the world and contributes nearly 80.0 million tons of fishes to the global production (FAO 2018). Currently, 152 million tons of food fishes is being used for human consumption where more than 50% is coming from aquaculture. Globally, the fish consumption per capita rapidly increased from 9.9 to 20.3 kg between 1960 and 2016 (FAO 2018). Moreover, fishes are one of the most-traded food commodities worldwide and contribute significantly to food security and adequate nutrition supply for the increasing population (FAO 2016).

However, the sustainable development of this rapidly growing sector is one of the major concerns particularly in terms of supply of safe and edible fishes. Aquaculture farming practices are changing worldwide from traditional to intensive farming systems, leading to stock a greater number of fish/shellfishes in lesser spaces of water, noticeably

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increasing the risk of transmissible and infectious diseases (Santos and Ramos 2018). In addition, aquaculture farming mainly depends on artificial feeds and sometimes overuse, misuse, or overfeeding of feeds are common scenarios due to lack of farmer's proper knowledge and institutional education specifically in developing countries. Subsequently, the culture environment changed from its natural to obnoxious condition which enhances the chances of disease burden, particularly infectious diseases. Diseases are considered as the single most impediment of aquaculture industry; in general, fishes are reared in congested and stressful environment which are more susceptible to bacterial infection, resulting a significant production loss (Santos and Ramos 2018). For the reduction of disease burden, the applications of antibiotics in prophylactic and therapeutic purposes are very common practices to prevent bacterial infections in aquaculture (Cabello et al. 2013). Prophylactic or therapeutic applications of antibiotics in aquaculture may exert the selection pressure to the natural bacterial population and enhance the ability to produce antibiotic-resistant bacteria or resistance genes to the aquaculture environment. Bacterial resistance to antibiotics is presently considered one of the most critical threats for human health and affecting the ability to treat a range of infections worldwide (WHO 2018; Hossain et al. 2020). Antibiotic resistance has already reached in upsetting levels in many countries globally, and some of the countries are currently using their last resort antibiotics for the treatment of bacterial infection.

Data related to antibiotic usage in aquaculture is very scarce and differs significantly from country to country. There is little to no enforcement/regulatory framework regarding antibiotic usage is present in many of the major aquaculture-producing countries particularly in Asia (Santos and Ramos 2018; Pruden et al. 2013; Chuah et al. 2016). Therefore, this review includes the current antibiotic uses patterns in aquaculture, the ecological, bacterial resistance, and human health risks of antibiotics, and the current regulatory framework and future concern of antibiotic use in aquaculture worldwide.

Application of antibiotics in aquaculture

Positive effects of antibiotics

The aquaculture farmers are regularly using antibiotics as prophylactic or therapeutic purposes to minimize the occurrence and spread of bacterial infections, particularly in the countries where no alternative preventive methods being implemented. Antibiotics are commonly used in aquaculture to prevent and/or cure infectious diseases in Asia, Canada, Europe, the USA, and many other countries worldwide (Tuševljak et al. 2013; Lulijwa et al. 2020; Preena et al. 2020). Usually, antibiotics are routinely applied in (i) prophylactic use by bath treatments or mixed with feed and (ii) therapeutic use for the treatment of bacterial infections (Ali et al. 2016). In addition, antibiotics particularly oxytetracycline and florfenicol are also used as growth promoter of the aquaculture species (Reda et al. 2013).

Side effects of antibiotics

In aquaculture, antibiotics are generally administered to the entire population including sick, healthy, and carrier individuals. As a result, antibiotics are mostly overused and misused in aquaculture of many countries (Santos and Ramos 2018). Moreover, antibiotic doses in aquaculture can be consistently higher than that of terrestrial animals farming, although the specific contamination levels are difficult to determine (Romero et al. 2012).

The application of antibiotics in finfish and shellfish aquaculture has often been questioned over the last decade in Asian countries because of lacking institutional education, training, and provision of disease diagnosis (Faruk et al. 2005; Holmström et al. 2003; Pham et al. 2015; Pathak et al. 2000). Due to the lack of institutional knowledge, the farmers usually overuse or misuse the antibiotics in the Asian aquaculture. Consequently, aquaculture products from different countries (Bangladesh, China, India, Malaysia, and Vietnam) have been rejected by the EU and the USA for the presence of banned antibiotics, e.g., chloramphenicol, nitrofuran, and furazolidone (FDA 2018). In addition, the use of excessive antibiotics in aquaculture might speed up the resistance development in aquaculture environment and cultured fishes. Furthermore, consumption of fish treated with antibiotics may cause deleterious effects on human health (Limbu et al. 2021).

The use of antibiotics in aquaculture may contaminate the culture environments as well as farmed organisms through different ways where feed is one of the most important sources (Li et al. 2021). A schematic diagram of antibiotic contamination from various probable sources in aquaculture is shown in Fig. 1.

Country-specific use of antibiotics in aquaculture

Though antibiotics are widely used in aquaculture, the data are still very limited. The information related to antibiotic usages and residues in aquaculture products mostly reported from developed countries; however, the majority of fish and shellfishes are being produced in developing and/or least developed Asian countries where the regulatory strategies or guideline in aquaculture are limited or almost absent (Howgate 1998; Chuah et al. 2016; Santos and Ramos 2018). Moreover, the application of antibiotics in aquaculture often lack institutional training and knowledge concerning

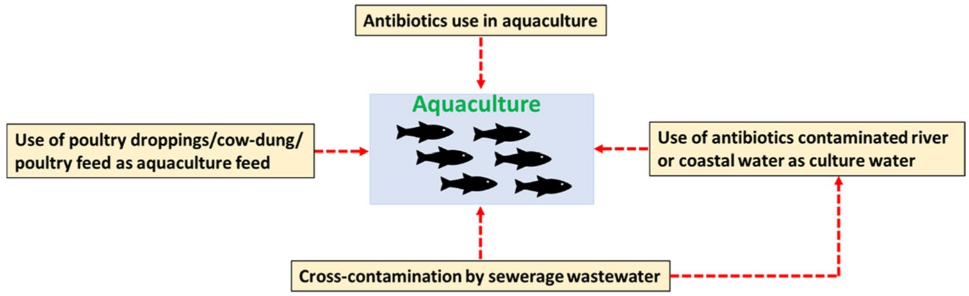

Fig. 1 Schematic diagram of probable sources of antibiotic contamination in aquaculture

the safe and responsible use of antibiotics (Gräslund et al. 2003; Pham et al. 2015), causing an unnecessary usage that consistently goes unstated. The major groups of antibiotics used in aquaculture are enlisted in Table 1 and the numbers of antibiotics used in different countries are shown in Fig. 2.

A recent review article reported several aspects of antibiotic usage in fifteen top fish-producing countries in the world (Lulijwa et al. 2020). Seventy-three percent of these countries used oxytetracycline, sulfadiazine, or florfenicol, while 55% of them applied sulfadimethoxine, erythromycin, amoxicillin, and enrofloxacin in aquaculture. The report also stated that 67 different types of antibiotics was applied in eleven of the fifteen countries from 2008 to 2018. Bangladesh (21), China (33), and Vietnam (39) were mentioned as the top three users of antibiotics in aquaculture. Reviewing the literature, we recorded 37 antibiotics; however, Lulijwa et al. (2020) reported thirty-nine antibiotics used in aquaculture of Vietnam. Some banned antibiotics are still being used in fish and shrimp aquaculture in Vietnam (Luu et al. 2021). Besides these three countries, other major fish-producing countries have been using various types of antibiotics in aquaculture shown in Fig. 2 and details in Table 1.

Country-specific data on antibiotic usage for the top fifteen aquaculture-producing countries were also reported previously by Sapkota et al. (2008). However, the application of antibiotics has recently been declined in Norway and this was due to the mandatory vaccination in aquaculture. The review also reported 12 different types of generic and 25 trademarked antibiotics authorized for use in salmonid aquaculture in Chile. The antibiotics used in the salmon aquaculture has been increased around 56% between 2005 and 2015 in Chile (Lozano et al. 2018). The Food and Drug Administration (FDA) has approved only four antibiotics to use in US aquaculture against pathogenic bacteria (Lozano et al. 2018). Chuah et al. (2016) reviewed information related to antibiotics used in catfish aquaculture. They found twenty-four antibiotics of different classes applied in catfish aquaculture worldwide.

In Asia, approximately 36 different antibiotics have been used in aquaculture in the seven major fish-producing countries (Bangladesh, China, India, Indonesia, Philippines, Thailand, and Vietnam) (Rico et al. 2012). The most widely applied antibiotic classes were reported as sulfonamides with trimethoprim, tetracycline, and quinolones. In all seven countries referred to earlier, oxytetracycline and chloramphenicol have been used in aquaculture; however, the majority of Asian countries have recently banned the use of chloramphenicol in fish farming. The highest number of reported antibiotics used in aquaculture was 31 and 17 antibiotics in Vietnam and China, respectively (Rico et al. 2012). Twenty antibiotics of eight different classes have been reported in the case of Chinese aquaculture (Liu et al. 2017). Of the 20 recorded antibiotics, some antibiotics have been developed as human medicine. They also reported the occurrence of 32 antibiotics in Chinese aquatic products belonging to six categories. The number of detected antibiotics surpassed both the antibiotics registered and approved for aquaculture in China (Liu et al. 2017). In Malaysia, 23 antibiotics of different classes were detected in aquaculture (Thiang et al. 2021).

In Bangladesh, a broad variety of antibiotics are applied in aquaculture as well as in larvae rearing in captive and nursery conditions (Ali et al. 2016; Rico et al. 2013; Shamsuzzaman and Biswas 2012). Previous research has shown that primarily seven antibiotics are commonly used in finfish and shellfish farms against the treatment of infectious diseases in Bangladesh (Ali et al. 2016). Aquaculture farmers confirmed that 77% antibiotics are used with fish feed, while the remaining 23% are used directly on surface water of the culture pond. Oxytetracycline and chlortetracycline were the most common antibiotics used in Bangladeshi aquaculture followed by amoxicillin, cotrimoxazole, doxycycline, sulfadiazine, sulfamethoxazole, and trimethoprim (Hossain et al. 2017). Presently, lesser amounts of antibiotics are being used in Bangladesh particularly in pangas or tilapia farming compared to that of other Asian countries (Rico et al. 2013).

Antibiotic-resistant bacteria and resistance genes in aquaculture

Aquaculture has been considered as a "genetic hotspot" for resistance gene transfer. Multiple antibiotic-resistant strains are now being frequently detected in fish/shellfish

Table 1 Antibiotic use in major aquaculture-producing countries in the world

Country name	Major groups of antibiotic use in aquaculture							Reference
	Aminoglycosides	Beta-lactams	Macrolides	Sulfonamides	Tetracycline	Quinolones	Others	
Bangladesh	NEO, KAN	AMX, PEN	ERY, TYL	SDZ, SMX, SMZ, TMP, TBS, CMZ, SMZL	CTC, DC, OTC	CIP	CHP, FZD, FRD, NFT	Sheikh et al. (2009); Rico et al. (2013); Faruk et al. (2008); Ali et al. (2016); Shamsuzzaman and Kumar (2012); Hossain et al. (2017); Hossain et al. (2018); Ali et al. (2018)
Brazil					TC, OTC	ENF	FLF	Guidi et al. (2018); Orlando et al. (2016); Monteiro et al. (2015); Monteiro et al. (2016)
Chile	NEO, CN	AMX	ERY	SDZ, SMZ, SMX, SMM, TMP	OTC, DC	OLA, FMQ, ENF, NRF	FLF, FZD, CHF, THP	Lozano et al. (2018); Miranda et al. (2018); Cabello et al. (2013); Liu et al. (2017)
China	CN, NEO	AMX, LEX, RAD, CTX	ERY, AZM, CLR, ROX	SCP, SDZ, SMX, SSZ, SDMX, SMZ, SDM, SMM, TMP, SPD, SQX	TC, CTC, OTC	CIP, ENF, NRF, OFL, ENC, SRF	LCM, FLF, THP	He et al. (2012); Chen et al. (2015); Liu et al. (2018a, 2018b); Chen et al. (2018); He et al. (2016); Rico et al. (2013); Song et al. (2016); Zheng et al. (2012); Zou et al. (2011); Luo et al. (2011); Zhong et al. (2018); Liu et al. (2018a, 2018b)
Egypt						CIP	FLF	Rezk et al. (2015)
India			ERY	SDZ, SMZS, SMX, SDM, SMZ, SPD, SMPD, SDX, SDTZ, SNM, STZ			CHP	Palaniyappan et al. (2013); Swapna et al. (2012)
Indonesia	NEO, STM		ERY		OTC	ENF	CHP, PFR	Supriyadi and Rukyani (2000)
Italy			AMX	SDZ/TMP	TC, OTC	FMQ		Labella et al. (2013)
Japan		AMP, AMX, PEN	ERY, SPC	SDMX	OTC	OLA, FMQ, NDA, MXC	FLF, THP	Sapkota et al. (2008)
Norway					OTC	OLA, FMQ	FLF	Midtlyng et al. (2011); Zhang et al. (2018)
Malaysia			AZM, TYL, ERY, CLR, ROX	SDZ, TBS, SPD, STZ, SMZS, SMZ, SMZL, SMX, SDMX, TMP	MNC, OTC, TC, DC	CIP, ENF, OFL, NOR, NDA	LCM	Thiang et al. (2021)
Philippines	NEO, KAN	AMX	ERY	SDZ, SDMX, SMM, TMP		ENF, NRF, OLA	FLF, NPN, RP	Cruz-Lacierda et al. (2008); Sonia et al. (2012)
S. Korea		AMX	ERY	SCP, SDZ, SMX, SDMX, SMZ, TMP, OTP	TC, CTC, OTC	CIP, ENF, OLA, NDA	FLF	Shim et al. (2010); Kang et al. (2018); Kim et al. (2010)
Sri Lanka		AMX		SDZ/TMP	TC, OTC	OLA, FMQ, SRF	FLF	Manage (2018)
Taiwan			ERY	SDMX	OTC	OLA, FMQ	CHP, FLF, FZD	Liao et al. (2000); Sapkota et al. (2008)
Thailand		AMX, PEN		SDZ, SDMX, SMM, TMP, OTP, TBS, SDMX+TMP, SGD	TC, OTC	ENF, NRF		Rico et al. (2013); Baoprasertkul et al. (2012); Rico et al. (2014)

Table 1 (continued)

Country name	Major groups of antibiotic use in aquaculture							Reference
	Aminoglycosides	Beta-lactams	Macrolides	Sulfonamides	Tetracycline	Quinolones	Others	
UK		AMX		CMZ	OTC	OLA, SAF		Alderman and Hastings (1998)
USA				Sulfa/TMP, SDMX, OTP	OTC		FLF	Dawood et al. (2017)
Vietnam	CN, NEO, KAN, APR	AMX, PEN, AMP, LEX	ERY, CLR	SDZ, SMX, SDMX, SMZ, SDM, TMP, OTP	TET, CTC, DC, OTC	CIP, ENF, NRF, OLA, FMQ, LVF, OFL	CHP, FLF, THP, FTD, NFZ, CT, RM, RP, MNZ	Huynh (2009); Phan et al. (2009); Phu et al. (2016); Thuy and Loan (2011); Hoang et al. (2012); Pham et al. (2015); Rico et al. (2013); Bosma et al. (2009); Thai et al. (2018)

Aminoglycosides: apramycin (APR), gentamycin (CN), kanamycin (KAN), neomycin (NEO), streptomycin (STM); *beta-lactams*: amoxicillin (AMX), penicillin (PEN), ampicillin (AMP), cephalexin (LEX), cephradine (RAD), cefotaxime (CTX); *macrolides*: erythromycin (ERY), azithromycin (AZM), clarithromycin (CLR), roxithromycin (ROX), spiramycin (SPC), tylosin (TYL); *sulfonamides*: sulfachloropyridazine (SCP), sulfadiazine (SDZ), sulfamerazine (SMZS), sulfamethoxazole (SMX), sulfisoxazole (SSZ), sulfadimethoxine (SDMX), sulfamethazine (SMZ), sulfadimidine (SDM), sulfamonomethoxine (SMM), trimethoprim (TMP), sulfapyridine (SPD), sulfamethoxypyridizine (SMPD), sulfadoxine (SDX), sulfadimethiazole (SDTZ), sulfanilamide (SNM), sulfathiazole (STZ), ormethoprim (OTP), tribrissen (TBS), sulfadimethoxine + trimethoprim (SDMX + TMP), cotrimoxazole (CMZ), sulfamethizole (SMZL), sulfaguanidine (SGD), sulfaquinoxaline (SQX); *tetracycline*: tetracycline (TET), chlortetracycline (CTC), oxytetracycline (OTC), doxycycline (DC), minocycline (MNC); *quinolones*: ciprofloxacin (CIP), enrofloxacin (ENF), norfloxacin (NRF), oxolinic acid (OLA), flumequin (FMQ), levofloxacin (LVF), nalidixic acid (NDA), ofloxacin (OFL), enoxacin (ENC), sarafloxacin (SAF), miloxacin (MXC); *others*: rifamycin (RM), rifampicin (RP), lincomycin (LCM), chloramphenicol (CHP), forazine/thorazine (FZ/TRZ), florfenicol (FLF), thiamphenicol (THP), colistin (CT), furazolidon (FZD), furaltadone (FTD), nitrofurazone (NFZ), nifurpirinol (NPN), nitrofurantoin (NFT), furadin (FRD), prefuran (PFR), metronidazole (MNZ)

and aquaculture environment, which greatly threat the medical treatment options as well as increase the unwanted deaths (Watts et al. 2017). Reverter et al. (2020) reported that aquaculture of low- and middle-income countries contributed the higher levels of antibiotic-resistant bacteria. Several previous studies and reviews confirmed the higher intensities of antibiotic-resistant bacteria and resistance genes isolated from aquaculture environments worldwide (Tomova et al. 2015; Kathleen et al. 2016; Santos and Ramos 2018). Antibiotic-resistant bacteria and resistance gene can be transmitted to other bacteria or organisms through horizontal or vertical gene transfer. Consequently, the whole population might be contaminated by antibiotic-resistant genes or bacteria in the aquaculture environment (Ruzauskas et al. 2018; Preena et al. 2020).

Antibiotic-resistant bacteria and resistance genes/plasmids in major aquaculture-producing countries are shown in Table 2. Antibiotic-resistant bacteria and resistance genes are widely detected in all of the major aquaculture-producing countries worldwide. The frequently isolated antibiotic-resistant bacteria are *Vibrio* spp., *Aeromonas* spp., *Bacillus* spp., *Pseudomonas* spp., *Enterobacteriaceae*, *Streptococcus* spp., *Exiguobacterium* spp., etc. Other bacteria species including *Flavobacterium* are also detected in the aquaculture environments (Strepparava et al. 2013). Some of the most frequently detected antibiotic resistance genes along with their respective antibiotic classes are tetracycline (*tetA*, *tetB*, *tetK*, *tetM*), quinolone (*qnrA*, *qnrB*, *qnrS*), sulfonamides (*sulI*), and others (details in Table 2). Some resistant bacteria and resistance genes are commonly detected in aquaculture, which are pathogenic to fish/shellfish and for human health (Guo et al. 2020; Shen et al. 2020). Fu et al. (2020) have recently reported that probiotics used in aquaculture contain antibiotic-resistant pathogenic bacteria particularly *Klebsiella pneumoniae* which may pose serious threat to culture organisms and human health. In the Southern USA, the gram-negative bacteria were isolated from antibiotic-treated aquaculture ponds and showed higher resistant to ampicillin, chloramphenicol, nitrofurantoin, oxytetracycline, and tetracycline compared to that from the untreated rivers (Kerry et al. 1996). Previous studies isolated 292 bacterial strains from salmon aquaculture and conducted a susceptibility test against quinolones, florfenicol, and oxytetracycline (Henríquez et al. 2016; Saavedra et al. 2017). These studies revealed a higher occurrence of resistance to quinolones, florfenicol, and oxytetracyclines in Chilean salmon aquaculture. Most of the salmon aquaculture farmers applied oxytetracycline and florfenicol in their aqua-farms to control or prevent *Piscirickettsia salmonis* (Henríquez et al. 2016; Saavedra et al. 2017; Du et al. 2019). The doses of florfenicol were almost 2-folds higher in 2016 than that of 2013, indicating that bacteria in aquaculture environment were getting resistant over the time against

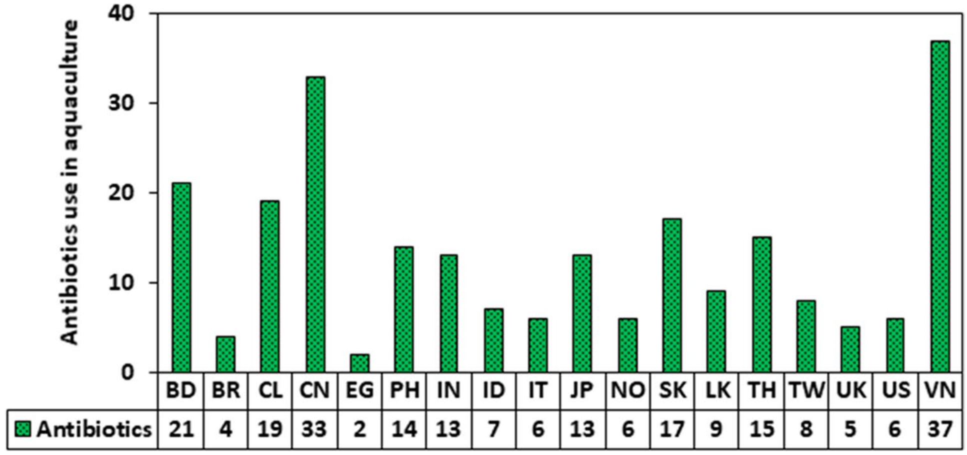

Fig. 2 Numbers of antibiotic use in major aquaculture-producing countries. BD, Bangladesh; BR, Brazil; CL, Chile; CN, China; EG, Egypt; PH, Philippines; IN, India; ID, Indonesia; IT, Italy; JP, Japan; NO, Norway; SK, South Korea; LK, Sri Lanka; TH, Thailand; TW, Taiwan; UK, United Kigdom; US, United States of America; VN, Vietnam

this specific antibiotic. Rozas and Enríquez (2014) reported that *P. salmonis* showed higher degree of resistance against tested antibiotics than that of earlier isolates.

Cross-contamination of antibiotics and other drugs in aquaculture

Aquaculture environment was mainly contaminated through the direct introduction of antibiotics for the disease prevention and production accelerating of farmed fishes and indirect transmission from improper use of veterinary and human antibiotics. The emerging contaminants (antibiotics and other drugs) originating from different anthropogenic activities adjacent to the farm areas may also have impacts on aquaculture environment. In general, river or coastal water is used for aquaculture purposes in most of the cases beside the underground water. Therefore, cross-contamination of contaminants is a very common scenario; however, data regarding this matter is very limited worldwide. A study has shown the probable effect of river waters on the aquaculture ponds which is contaminated by anticancer drugs (220 ng/L) (Lai et al. 2018). The anticancer drugs typically used in hospitals or nursing homes or drug production facilities, and households, however, detected in higher concentration in aquaculture pond water would noticeably be indicative of cross-contamination from adjacent waterbodies. The other drugs such as methadone, methamphetamine, and ketamine were measured in aquaculture water at concentrations of 0.3–22.7 ng/L, whereas methadone was detected most frequently (92%) in aquaculture areas (Lai et al. 2018). The findings from these studies clearly indicated that cross-contamination by human drugs from the adjoining waters might have a great impact on aquaculture and escalated the risk to human health and aquatic organisms. Conversely, the antibiotic residues in wastewater of aquaculture being released into the natural water environments may pose a potential ecological risk to the aquatic organisms (Rico et al. 2013).

Risks assessment

Ecological risk of antibiotics

Antibiotic residues in aquaculture environment may have several adverse ecological impacts reported in earlier researches (Luo et al. 2011; Sun et al. 2016). Generally, researchers evaluate the ecological impacts based on measured or predicted concentration of antibiotics in the environment. However, it might be underestimated because significant amount of applied antibiotics is degraded through hydrolysis, photo-oxidation, and/or microbial action in aquatic environments (Yamamoto et al. 2009; Batchu et al. 2014; Vila-Costa et al. 2017) and some antibiotics are accumulated in aquatic organisms (Sangion and Gramatica 2016; Moreno-González et al. 2016).

In general, ecological risk is calculated by the following Eq. (1).

$$\mathrm{RQs} = \frac{\mathrm{PECorMEC}}{\mathrm{PNEC}} \tag{1}$$

where.

RQs is the risk quotients.

PEC = Predicted environmental concentration.

MEC = Measured environmental concentration.

PNEC = Predicted no-effect concentration.

The predicted no-effect concentration (PNEC) is computed by the following Eq. (2).

$$\mathrm{PNECs} = \frac{\mathrm{LC_{50}orEC_{50}}}{\mathrm{AF}} \tag{2}$$

Table 2 Antibiotic-resistant bacteria and resistance genes in major aquaculture-producing countries in the world

Country	Resistant bacteria	Resistant antibiotics	Reference	Resistance genes/plasmid	Antibiotic class	Recipient bacteria	Reference
Australia	*Vibrio* spp. and *Aeromonas* spp.	Ampicillin, amoxicillin, cephalexin, erythromycin, oxytetracycline, tetracycline, nalidixic acid, sulfonamides, chloramphenicol, florfenicol, ceftiofur, cephalothin, cefoperazone, oxolinic acid, gentamicin, kanamycin and trimethoprim	Akinbowale et al. (2007)				
Bangladesh	*V. parahaemolyticus*	Ampicillin, chloramphenicol, erythromycin, gentamicin, nitrofurantoin, oxolinic acid, tetracycline, ciprofloxacin, polymyxin- B	Haque et al. (2015)				
	Aeromonas	Ampicillin, erythromycin, cephalothin, nalidixic acid, streptomycin, tetracycline, chloramphenicol, and gentamicin	Rahman et al. (2009)				
Brazil	*Vibrio* sp.	Ampicillin and tetracycline	Reboucas et al. (2011)				
Canada	*Flavobacterium psychrophilum*	Florfenicol, oxytetracycline, trimethoprim–sulfamethoxazole, and ormetoprim–sulfadimethoxine	Hesami et al. (2010)				
Chile	*Bacillus aryabhattai, Exiguobacterium sibiricum, Marinobacter litoralis, Psychrobacter pulmonis, Stenotrophomonas maltophilia, Thalassospira xiamenensis*	Tetracycline and sulfonamides	Shah et al. (2014)	*floR*	Florfenicol	*Pseudoalteromonas* sp., *Shewanella* sp., *Cobetia* sp., *Marinobacter* sp., *Halomonas* sp.	Tomova et al. (2015)

Table 2 (continued)

Country	Resistant bacteria	Resistant antibiotics	Reference	Resistance genes/plasmid	Antibiotic class	Recipient bacteria	Reference
				qnrA, qnrB, qnrS, aac(6')1b	Quinolone	*Pseudomonas* sp., *Alcanivorax* sp., *Arcobacter* sp., *Arthrobacter* sp., *Kytococcus* sp., *Marinobacter* sp., *Microbacterium* sp., *Rhodococcus* sp., *Actinobacterium* sp., *Cellulophaga* sp., *Flavobacteriaceae*, *Erythrobacter* sp., *Tsukamurella* sp., *Dietzia* sp., *Microbacter* sp.	
				aac(6')-Ib-cr	Quinolone	*Sporosarcina* sp., *Rhodococcus* sp., *Kytococcus* sp., *Erythrobacter* sp.	Aedo et al. (2014)
				tetA, tetB, tetK, tetM	Tetracycline	*Pseudoalteromonas* sp., *Shewanella* sp., *Psychrobacter* sp., *Cobetia* sp., *Vibrio* sp., *Pseudomonas* sp.	
China	*Vibrio splendidus, Vibrio tasmaniensis, Pseudoalteromonas marina, Mucus bacterium, Pseudoalteromonas haloplanktis, Aeromonas* sp.	Oxytetracycline, streptomycin, chloramphenicol, ampicillin, nalidixic acid, multiple antibiotics	Dang et al. (2007); Deng et al. (2016)	*qnrVC4*	Quinolone	*Aeromonas punctata*	Xia et al. (2010)
Europe	*Pseudomonas, Aeromonas, Chryseobacterium, Enterobacteriaceae*	β-Lactams and others	Ruzauskas et al. (2018)				
India	*Enterobacteriaceae* and *Flavobacterium, Aeromonas* sp.	Oxytetracycline, ampicillin, oxytetracycline, amoxicillin, novobiocin	Singh et al. (2009); John and Hatha (2012)				
Japan	*Streptococcus dysgalactiae*	Macrolide and tetracycline	Nguyen et al. (2017)	*tetB, tetY, tetD*	Tetracycline	*Photobacterium* sp., *Vibrio* sp., *Alteromonas* sp.	Furushita et al. (2003)
	Edwardsiella tarda	Tetracycline, streptomycin and kanamycin	Yu et al. (2012)	*mef(C), mph(G)*	Macrolide	*P. damselae* subsp. *damselae*	Nonaka et al. (2012)
	Photobacterium, Vibrio sp., *Pseudomonas, Alteromonas, Citrobacter, and Salmonella* spp., *Vibrio* sp.	Tetracycline, oxytetracycline	Furushita et al. (2003); Nonaka and Suzuki (2002)				

Table 2 (continued)

Country	Resistant bacteria	Resistant antibiotics	Reference	Resistance genes/ plasmid	Antibiotic class	Recipient bacteria	Reference
	Pseudomonas aeruginosa	Tetracycline, ampicillin, and erythromycin	Lamari et al. (2017)				
	Photobacterium damselae	Kanamycin, chloramphenicol, tetracycline, and sulfonamide	Kim et al. (2008)				
Norway	*Flavobacterium psychrophilum*	Quinolones	Shah et al. (2012)				
Philippines	*Vibrio harveyi*	Oxytetracycline, oxolinic acid, chloramphenicol, furazolidone	Tendencia and Pena (2001)				
France				*qnrA*	Quinolone	*S. algae*	Poirel et al. (2005)
				qnrS2	Quinolone	*A. punctata* subsp. *punctata, Alloteuthis media*	Cattoir et al. (2008)
				mcr-1	Polymyxin	*Shewanella algae MARS 14*	Telke et al. (2015)
Spain				*ICEVspPor2, ICEvalPor1*	Rifampicin	*Vibrio splendidus,V. alginolyticus,Shewanella haliotis,Erythrobacter nigricans*	Rodríguez-Blanco et al. (2012)
South Africa	*Aeromonas salmonicida* and *A. hydrophila*	Tetracycline and amoxicillin	Jacobs and Chenia (2007)				
S. Korea	*Aeromonas hydrophila, A. allosaccharophila, A. veronii, A. sobria, A. caviae,* and *A. media*	Quinolones	Chenia (2016)	*AadA, tetB, tetD, blaTEM*	Aminoglycoside, Tetracycline, β-Lactam	*Escherichia coli*	Ryu et al. (2012)
Taiwan	*Streptococcus dysgalactiae*	Macrolide and tetracycline	Nguyen et al. (2017)				
Thailand	*Aeromonas veronii* and *A. aquariorum*	Tetracycline and ampicillin	Yano et al. (2015)				
Turkey				*floR, sulI, tetC, tetD, tetE*	Florfenicol, Sulfonamides, Tetracycline	*Yersinia ruckeri, Photobacterium damselae*	Duman et al. (2017)

Table 2 (continued)

Country	Resistant bacteria	Resistant antibiotics	Reference	Resistance genes/plasmid	Antibiotic class	Recipient bacteria	Reference
USA	*Photobacterium damselae*	Kanamycin, chloramphenicol, tetracycline, and sulfonamide	Kim et al. (2008)	*floR, blaCMY-2*	Florfenicol, Beta-lactam	*Edwardsiella ictaluri*	Welch et al. (2009)
	Aeromonas hydrophila and *Plesiomonas shigelloides*	Tetracycline, oxytetracycline, ampicillin, chloramphenicol, kanamycin, nitrofurantoin	McPhearson et al. (1991)				
Vietnam	*E. coli*	Ampicillin, ciprofloxacin, tetracycline, sulfafurazole, gentamicin, cephalothin, amoxicillin/clavulanic, chloramphenicol, enrofloxacin, kanamycin, nalidixic acid, norfloxacin, streptomycin, trimethoprim	Van et al. (2008)	*aadA, aphA-1, cat1, cmlA, dhfrV, sulI, tetA, tetB, TEM*	Aminoglycoside, Chloramphenicol, Trimethoprim, Sulfonamide, Tetracycline, Beta-lactam	*E. coli*	Van et al. (2008)

where LC_{50} or EC_{50} is the obtained lowest median effective concentration and AF means assessment factor.

The VICH (2004) incorporated the ecological risk assessment of antibiotics being used in aquaculture. A recent study on the ecological and human health risks of antibiotics has been conducted for aquaculture in China (Sun et al. 2016). A toxicity database (e.g., www.epa.gov/ecotox) and published literature were used to collect data for primary producers, invertebrates, and fish in their study. Sun et al. (2016) have prioritized taking the toxicity data on cyanobacterial species (*Microcystis aeruginosa*) for PNEC calculations. Toxicity values for *Chlorella vulgaris* or *Scenedesmus subspicatus* were considered in case of data unavailability on cyanobacteria toxicity. For invertebrates (e.g., *Daphnia magna*), they considered EC_{50} as the acute endpoints to calculate the PNECs, where the value for EC_{50} was obtained for the immobilization or mortality upon exposure from 1 to 4 days. The LC_{50} was attained by considering the acute endpoints of fish while the exposure period was 1 to 7 days. The lowest EC_{50} value was taken into account when different values were available for the same organism. The PNEC was calculated for ecological risk analysis in dividing the attained EC_{50}/LC_{50} data by an assessment factor (AF) of 100 (VICH 2004). Previously, four antibiotics such as ampicillin, levofloxacin, sarafloxacin, and sulfadiazine have been shown to have an adverse ecological effect to primary producers, while no acute risk was found for other aquatic organisms. Several antibiotics showed no adverse acute risk to sensitive aquatic organisms; however, they are predicted to have long-term effects on aquatic organisms.

Another approach—the risk quotient (RQ) approach—was adopted by many environmental scientists to assess the environmental risk of antibiotics by the following EC guidance (EC 2003). This method has been applied mainly to evaluate the risk of antibiotic compounds in rivers, streams, bays, or lakes. This approach may also be applicable in ecological risk assessment of the waterbodies where fish are farmed in pond, cage, or pen. Generally, the measured environmental concentration (MEC) and PNEC were used to compute the RQ values. The PNEC is obtained from toxicity data (EC_{50} or LC_{50}) or from NOEC values divided by an assessment factor (i.e., 1000, 100, or 10). In the absence of data on ecotoxicity, the ECOSAR model is used to get the toxicity data of three trophic levels (algae, *Daphnia*, and fish). The available toxicity data of antibiotic compounds are included in Table 3.

EC (2003) classified ecological risk as three categories: the minimum risk $RQ < 0.1$; the median risk $RQ < 1$; and the highest risk $RQ \geq 1$ (EC 2003; Sanderson et al. 2003; Hernando et al. 2006). Risk level $RQ < 0.1$ depicts no or minimal effects on growth and reproduction of aquatic organisms including fish; $RQ < 1$ may impact 50% of exposed population by hampering growth and reproduction whereas $RQ \geq 1$ may cause highest impacts by retardation of organism's growth and reproduction which may eventually lead to population decline. In Hong Kong, the RQ approach was used to evaluate ecological risk in river water by computing RQs where the MEC and PNECs of each antibiotic have been used for different aquatic organisms (Deng et al. 2016). They used the calculated RQ values to define the ecological risk into four categories (negligible risk: $RQ < 0.01$; low risk: $0.01 \leq RQ < 0.1$; medium risk: $0.1 \leq RQ < 1$; and high risk: $RQ \geq 1$). In their analysis, ofloxacin and doxycycline were found to pose definite ecological risk to algal population, whereas sulfonamides did not pose a significant ecological risk to the tested aquatic organisms. The other studies used the VICH and EC guidelines to calculate the ecological risk of different antibiotics measured from the surface water of aquaculture in Bangladesh (Hossain et al. 2017, 2018). In China, the ecological risk for ciprofloxacin, erythromycin, enrofloxacin, ofloxacin, and tetracycline was determined and showed medium to high risks to algae and bacteria in aquatic ecosystem (Chen et al. 2021).

Resistance risk of antibiotics

Apart from ecological risk of antibiotics in aquaculture, antibiotic resistance risk in the environment is of recent major concern for the scientists. A new approach has recently been applied for the antibiotic's resistance risk assessment by the following Eq. (3).

$$RQs = \frac{\mathrm{PECorMEC}}{\mathrm{PNECforresistanceselection}} \quad (3)$$

PNEC for resistance selection means that resistance is acquired when the concentration is higher than the value. In aquaculture, using the resistance data, some previous studies assessed the resistance risk of antibiotics by following the above equation (Bengtsson-Palme and Larsson 2016).

Multiple antibiotic-resistant bacteria and resistance genes were present in the aquatic ecosystem, and significant associations were documented for different antibiotics in natural waterbodies (Zheng et al. 2011). The water samples from Chinese aquaculture were previously analyzed with the target of 21 antimicrobials. The RQ approach was applied for the evaluation of the resistance risk for 14 antimicrobials. RQ values were obtained by dividing the measured concentration of antibiotics and PNECs of antibiotics for resistance selection (Sun et al. 2016). The authors reported that RQ values for 7 antimicrobials out of 14 were greater than 1 while the RQs were more than 10 for 3 antibiotics have shown.

A recent study conducted by Hanna et al. (2018) in China has shown that RQ values (resistance risk) for enrofloxacin and levofloxacin were higher than 1.0 and ranged from 0.1 to 1.0 for ciprofloxacin. Antibiotics in surface waters also

Table 3 Acute median effective concentrations LC_{50} or EC_{50} (mg/L) of selected antibiotics on aquatic organisms

Class	Antibiotics	Species	LC_{50}/EC_{50} (mg/L)	Reference
Quinolones	Norfloxacin	*S. obliquus*	50.18 (96 h)	Lu et al. (2007)
		L. gibba	0.913 (7 days)	Brain et al. (2004)
		D. magna	194.98 (48 h)	Lu et al. (2007)
	Ciprofloxacin	*M. aeruginosa*	0.017 (24 h)	Robinson et al. (2005)
		P. subcapitata	18.7 (24 h)	Robinson et al. (2005)
		L. minor	0.203 (24 h)	Robinson et al. (2005)
		L. gibba	0.698 (7 days)	Brain et al. (2004)
	Enrofloxacin	*M. aeruginosa*	0.049 (24 h)	Robinson et al. (2005)
		P. subcapitata	3.10 (24 h)	Robinson et al. (2005)
		L. minor	0.114 (24 h)	Robinson et al. (2005)
		D. magna	56.7 (48 h)	Park and Choi (2008)
		M. macrocopa	> 200 (48 h)	Park and Choi (2008)
		O. latipes	> 100 (48 h)	Park and Choi (2008)
	Ofloxacin	*M. aeruginosa*	0.021 (24 h)	Robinson et al. (2005)
		P. subcapitata	12.10 (24 h)	Robinson et al. (2005)
		P. subcapitata	1.44 (72 h)	Cunningham et al. (2006)
		L. minor	0.126 (24 h)	Robinson et al. (2005)
		L. gibba	0.532 (7 days)	Brain et al. (2004)
		C. dubia	17.41 (48 h)	Cunningham et al. (2006)
Sulfonamides	Sulfamethoxazole	*S. leopolensis*	0.027 (96 h)	Cunningham et al. (2006)
		L. gibba	0.081 (7 days)	Brain et al. (2004)
		L. gibba	0.21 (7 days)	Isidori et al. (2005)
		C. dubia	15.51 (48 h)	Cunningham et al. (2006)
		D. magna	177.3 (96 h)	Kim et el. (2007)
		O. latipes	562.5 (96 h)	Kim et el. (2007)
	Sulfamethazine	*S. vacuolatus*	9.52*	Białk-Bielinska et al. (2011)
	Trimethoprim	*R. salina*	16**	Lützhøft et al. (1999)
	Sulfadiazine	*M. aeruginosa*	0.135 (72 h)	Lützhøft et al. (1999)
		S. capricornutum	2.2 acute	Eguchi et al. (2004)
		D. magna	221 (48 h)	Wollenberger et al. (2000)
Macrolides	Roxithromycin	*L. gibba*	> 1 (7 days)	Brain et al. (2004)
		D. magna	7.1 (96 h)	Choi et al. (2008)
		M. macrocopa	39.3 (96 h)	Choi et al. (2008)
		O. latipes	288.3 (96 h)	Choi et al. (2008)
	Erythromycin	*P. subcapitata*	0.02 (72 h)	Cunningham et al. (2006)
		L. gibba	> 1 (7 days)	Brain et al. (2004)
		L. minor	5.62 (7 days)	Pomati et al. (2004)
		B. calyciflorus	0.94 (48 h)	Cunningham et al. (2006)
	Tylosin	*M. aeruginosa*	0.034	Halling-Sørensen (2000)
		D. magna	680**	Wollenberger et al. (2000)
	Roxithromycin	*P. subcapitata*	0.01*	Yang et al. (2008)
Tetracycline	Oxytetracycline	*M. aeruginosa*	0.207**	Lützhøft et al. (1999)
	Tetracycline	*M. aeruginosa*	0.09**	Halling-Sørensen (2000)
	Chlortetracycline	*M. aeruginosa*	0.05**	Halling-Sørensen (2000)

Brachionus calyciflorus = *B. calyciflorus; Brachydanio rerio* = *B. rerio; Ceriodaphnia dubia* = *C. dubia; Daphnia magna* = *D. magna; Lemna gibba* = *L. gibba; Lemna minor* = *L. minor; Microcystis aeruginosa* = *M. aeruginosa; Moina macrocopa* = *M. macrocopa; Oryzias latipes* = *O. latipes; Oncorhynchus mykiss* = *O. mykiss; Pseudokirchneriella subcapitata* = *P. subcapitata; Synechoccus leopolensis* = *S. leopolensis; Scenedesmus obliquus* = *S. obliquus*

*NOEC value

**Acute toxicity value

influence the risk of developing antimicrobial resistance and resistance genes, which may pose adverse impacts on humans and other organisms in the environment (Guo et al. 2020; Shen et al. 2020). The existence of residual antibiotics in the aquatic environment might facilitate the development of antibiotic resistance, thus enhancing the environmental risks of antibiotic resistance (Chen et al. 2021). Particularly, Hossain et al. (2017) investigated the occurrence of the commonly used antibiotics in surface waters of aquaculture and further measured the resistance risk of detected antibiotics. The results revealed a low risk (RQ < 1.0) for antibiotic resistance of the detected antibiotics. The observed and size-adjusted minimum inhibitory concentrations (MIC) of antibiotics and PNECs for resistance selection are presented in Table 4.

Human health risk of antibiotics

The human exposure to residual antibiotics presents in food and environmental media have been thought not probably to pose acute toxicity at low concentration; however, less are known regarding the chronic toxicity (Jones et al. 2004). Furthermore, there is another issue related to the understanding limitations on how the repeated exposures to low-level antibiotic concentrations might cause toxicity on individual aquatic organisms. Several studies reported that the widespread and continuous application of antibiotics was common in aquaculture in Thailand and Philippines (Gräslund et al. 2003; Holmström et al. 2003; Primavera et al. 1993).

Antibiotics are often misused and overused in livestock rearing and aquaculture farming, which might increase the antibiotic residues in food items (Liu and Wong 2013). Presently, more than 70 antibiotics have been detected in food items of animal origin including fish products (He et al. 2012). Recently, Lulijwa et al. (2020) reported the concentration of 14 antibiotics detected in fish and shellfish exceeded the country-specific MRL guidelines. The detected highest concentrations of antibiotics which exceeded the MLR values set by different countries are as follows: 1,801,400.0 ng/g of amoxicillin in *Labeo rohita* in Bangladesh (Ferdous et al. 2019); 1379.0 ng/g of oxytetracycline in Brazil (Monteiro et al. 2016); 200 ng/g oxytetracycline (Liu et al. 2018a, 2018b); 100.5 ng/g norfloxacin (He et al. 2016); and 15,090 ng/g of erythromycin (Chen et al. 2015) in China; 100.0 ng/g of sulfamethoxypyridine, sulfadimethoxine, and sulfamethazine (Palaniyappan et al. 2013) in India; 829.0, 500.0, 3341.0, 600.0, and 464.0 ng/g of amoxicillin, enrofloxacin, oxolinic acid, sulfadiazine, and trimethoprim in South Korea, respectively (Kang et al. 2018); and in Vietnam (196.0, 355.0, 149.0, 5800.0, and 1000 ng/g of ciprofloxacin, enrofloxacin, ofloxacin, sulfamethoxazole, and trimethoprim, respectively) (Uchida et al. 2016). The maximum residue limit exceeded in many countries (Fig. 3).

Consequently, humans can be exposed to different classes of antibiotics through the consumption of contaminated fish/shellfish or other aqua-foods and susceptible to adverse health risk (Chen et al. 2015). The risk associated with the exposure of antibiotic residues to human health may be expressed in two ways: (i) adverse drug reaction (ADR) such as allergic reaction and (ii) potential exposure to bacteria of clinically importance particularly antibiotic-resistant bacteria and resistance genes (Liu et al. 2017; Bousquet 2009). A great percentage of antibiotics such as sulfonamides, tetracycline, and penicillin G have antigenicity and consuming contaminated fish/shellfishes might enhance allergic reactions. Some antibiotics (e.g., quinolones and tetracyclines) used in aquaculture may impact on the progress of toddler's teeth (Kümmerer 2009). Some antibiotics have metabolites which are more toxic than the parents' drugs (Heuer et al. 2009). In addition, antibiotic's presence in water may pose human health risk (dermal cancer) through dermal contact (Lu et al. 2018).

Recently, the antibiotic resistances in aquaculture products have also shown a major health concern issue worldwide. The presence of antibiotics in aquaculture environment might speed up the development of resistant genes of bacteria strains, which may eventually transfer to humans through food chains. Multiple antibiotic-resistant bacteria were isolated from seafood in China (Broughton and Walker 2010; He et al. 2016), which might be transmitted to the human body through the consumption of raw seafood.

The Ministry of Agriculture and Rural Affairs of China in 2019 (GB 31,650–2019; effective from April 1, 2020) decided the maximum residue limits of antibiotics allowed to aquatic products as 1000 ng/g for florfenicol; 500 ng/g for neomycin; 300 ng/g for cloxacillin; 200 ng/g for erythromycin; 100 ng/g for enrofloxacin, ciprofloxacin, and total sulfonamides; and 50 ng/g for thiamphenicol, and no residual level or 0 ng/g is for chloramphenicol and furazolidone (Liu et al. 2017). In addition, Chen et al. (2017) have recently used in their study of the maximum residue limit for several antibiotics by following different guidelines of Japan, Brazil, Chile, FDA, and China (Table 5).

However, many antibiotics are yet not to be decided for their residual levels in fish muscle or skin. In addition, combine/additive concentrations of antibiotics in fish muscle or skin are not considered in the current regulation worldwide.

Current antibiotic policy orientation and regulatory frameworks

Antibiotics have been globally applied in aquaculture to prevent and/or treat bacterial infections. Consequently, the residues of antibiotics in aquaculture products may be unintentionally exposed to the human body through

Table 4 Size-adjusted lowest MIC and PNEC values (in μg/L) for resistance selection of antibiotic use in aquaculture. Source: adopted from Bengtsson-Palme and Larsson (2016)

Class	Antibiotics	Observed lowest MIC	Size-adjusted lowest MIC	PNEC for resistance selection	Covered genera (families)
Aminoglycoside	Gentamycin	16	16	1	27 (14)
	Kanamycin	125	32	2	10 (7)
	Neomycin	125	16	2	6 (5)
Beta-lactams	Amoxicillin	4	2	0.25	19 (12)
	Ampicillin	4	4	0.25	25 (15)
	Cephalexin	250	32	4	7 (5)
	Cefotaxime	2	1	0.125	19 (10)
	Benzylpenicillin	4	4	0.25	12 (11)
Macrolides	Azithromycin	16	4	0.25	6 (6)
	Clarithromycin	8	2	0.25	10 (10)
	Erythromycin	16	8	1	14 (13)
	Roxithromycin	32	8	1	7 (7)
	Tylosin	2000	32	4	1(1)
Sulfonamides	Sulfamethoxazole	1000	125	16	6 (4)
	Trimethoprim	16	8	0.5	15 (7)
Tetracycline	Doxycycline	32	16	2	20 (11)
	Oxytetracycline	125	4	0.5	2 (2)
	Tetracycline	16	16	1	30 (18)
Quinolones	Ciprofloxacin	2	1	0.064	29 (18)
	Enrofloxacin	8	0.5	0.064	4 (3)
	Flumequin	64	4	0.25	3 (2)
	Norfloxacin	16	4	0.5	12 (8)
	Levofloxacin	4	4	0.25	24 (16)
	Nalidixic acid	500	125	16	13 (5)
	Ofloxacin	8	4	0.5	20 (14)
Polymyxin	Colistin	64	16	2	10 (4)
Rifamycin	Rifampicin	2	0.5	0.064	12 (12)
Chloramphenicols	Thiamphenicol	500	8	1	1 (1)
	Florfenicol	125	16	2	6 (5)
	Chloramphenicol	125	64	8	18 (11)
Nitrofurans	Nitrofurantoin	4000	500	64	5 (4)
Lincosamides	Lincomycin	500	16	2	2 (2)
Nitroimidazoles	Metronidazole	16	2	0.125	3 (3)

ingestion and raise wide concerns (Liu et al. 2017; Wang et al. 2017). The farmed eels and eel products imported from China were repeatedly rejected from Japan during 1995–2000 because of higher concentration of antibiotics that exceeded the guideline values of Japanese government (FAO 2005). In 2016, the Chinese government strictly followed the regulation of antibiotic use in aquaculture, which might lower the antibiotic residues in aquatic products (Cui et al. 2017). Regarding present policies, the Food and Drug Administration (FDA), European Medicines Agency (EMA), the European Commission (EC), the Norwegian Food Safety Authority (NFSA), Codex Alimentarius Commission (Codex), and ministries of each country play the vital roles where EU, FDA policies, and regulations are widely used (Lulijwa et al. 2020). In the USA, the FDA monitor and collect data of the use of antibiotics in food animal production (FDA 2015). The antibiotics such as florfenicol, oxytetracycline, sulfamerazine, and sulfadimethoxine-ormetoprim are authorized by the FDA for aquaculture use (Chuah et al. 2016). Some antibiotics are authorized to use in aquaculture permitted by the regulation of the USA and EU member states (Mo et al. 2017). The application of antibiotics in aquaculture is highly regulated in developed countries. For example, FAO's Codex, the European Union (EU), EMA, and EC have been very persuasive in fixing up MRLs for antibiotics in food (FDA 2015; Codex 2017).

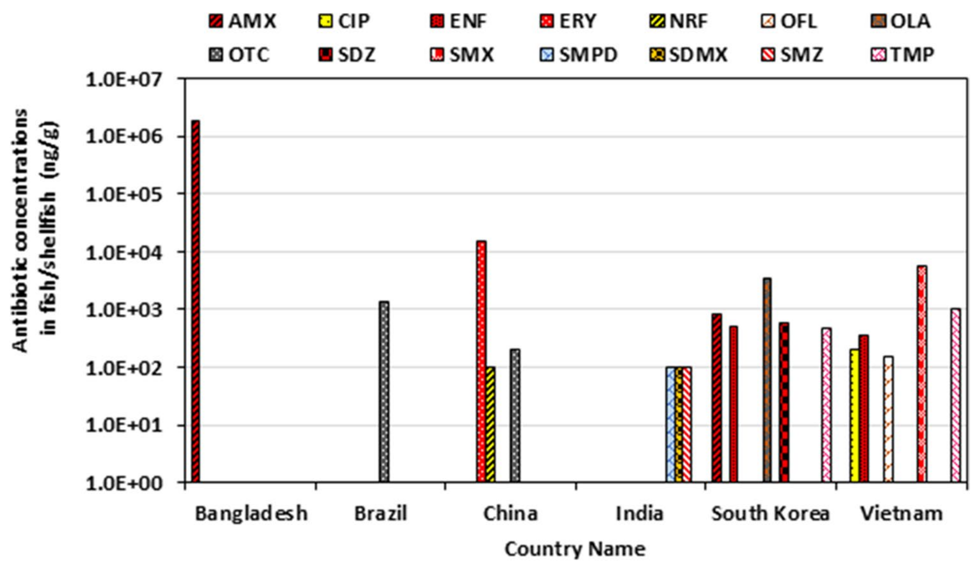

Fig. 3 The concentrations of antibiotics detected in fish/shellfish tissues. Here, we only considered the highest concentrations which exceeded the maximum residue limits (MRLs) of guideline values set by different countries

In EU region, the application of veterinary antibiotics is regulated by EU Council Regulations No. 37/2010 of 2009 (EC 2009a) and No. 470/2009 of 2009 (EC 2009b). These regulations are supported by the rigorous monitoring of the sale and use of antibiotics in EU countries by ESVAC (EMA 2018). However, the NFSA and Norwegian Veterinary Institute have proven the exceptional regulation mechanisms of antibiotic use and implementation in Norwegian aquaculture and NFSA regulates antibiotic use in aquaculture by the regulation No. 410/2009 (NFSA 2018). Use of antibiotics in Norwegian's aquaculture is strictly monitored and regulated by veterinarians and aqua-medicine biologists (Bailey and Eggereide 2020). The Veterinary Medicines Register (VetReg) was started by the Norwegian Food Safety Authority in 2011 to collect data of veterinary medicinal products (VMPs) prescribed for use in farmed fish. The Norwegian veterinary medicine (VMP) wholesalers and feed mills are instructed to report their sales to pharmacies and fish farmers to the Norwegian Public Health Institute (NPHI). The sales data of VMPs for use in farmed fish were annually published by NPHI (NVI 2016). In 2014, a total amount of only 511 kg (while 523 kg was prescribed) of antibiotics was used to produce about 1.3 million tons farmed fish in Norway. In the last decade, the use of antibiotics in producing salmon reduced from 1.0 to 0.36 mg/kg fish production in 2014. As a consequence, there was a very low probability of development of antibiotic resistance in Norwegian aquaculture and the transmission of such resistance to humans. Moreover, consumption of farmed fish produced in Norway might pose a negligible risk to human health in terms of antibiotic resistances (NVI 2016).

Suggestions for current policies and future thinking

Some approaches should be adopted globally on an immediate basis. Firstly, registration of aquaculture antibiotics must be obligatory in every country, and the limits of antibiotic use in aquaculture might be established. Moreover, the use of important and critically important antibiotics (WHO categorized) particularly human antibiotics should be prohibited to use in aquaculture since many countries have already started to use the last-resort antibiotics against bacterial infections. Furthermore, international scientists need to take alternative initiative to limit the development and spread of bacterial infections and bacterial resistances in aquaculture. Other probable approaches that should be ensured include such practices as good husbandry conditions and use of nutritious fish feed specifically in developing countries. In addition, the use of phage therapy against the bacterial infection in aquaculture might be helpful to reduce the burden of antibiotics and antibiotic resistances (Donati et al. 2021). Thus, a stringent policy or regulatory framework is urgently needed to be established giving superior emphasis on developing countries. The FDA, EC, EMA, NFSA, Codex, and respective ministries of government and other concerned authorities may play a vital role to make a common policy and regulation for the major aquaculture-producing countries (Lulijwa et al. 2020). In addition, one health conceptual framework might be considered for the surveillance of antimicrobial resistances in aquaculture (Santos and Ramos 2018). In addition, WHO/FDA/FAO or other similar authorities may work together to reduce the use of antibiotics for the minimization of potential risks. Finally, new technologies and/or management

Table 5 The maximum residue levels (MRLs) for antibiotics in animal tissues (sources: adopted from Chen et al. 2017; Lozano et al. 2018; Lulijwa et al. 2020)

Antibiotics	MRLs (ng/g)	Country[a]
Aminoglycoside		
Neomycin	500	China
Beta lactams		
Amoxicillin	50	China and South Korea
Ampicillin	50	China
Macrolides		
Erythromycin	200	China
Sulfonamides		
Sulfadiazine	100	South Korea
Sulfamethoxazole	20*, 100**	Japan*, Vietnam**
Sulfaquinoxaline	100	FDA
Sulfamethoxypyridine	100	India
Sulfadimethoxine	100	India
Sulfamethazine	100	India
Sulfonamides (all combined)	100	China
Trimethoprim	50	China, Japan, Vietnam, South Korea
Tetracycline		
Doxycycline	100	China
Oxytetracycline/chlortetracycline/tetracycline	200	China
Oxytetracycline	200*, 100**	Chile*, Brazil*, EU**
Quinolones		
Ciprofloxacin	200*, 100**	China*, Vietnam**
Enrofloxacin	100	China, Vietnam, South Korea
Ofloxacin	50*, 100**	Japan*, Vietnam**
Oxolinic acid	100	China and South Korea
Norfloxacin	100	China, Japan
Sarafloxacin	30	China

Symbols "*" and "**" are used for similarity in the same row

MRLs maximum residue levelsi

[a]The MRLs of antibiotics set by the Guideline of the Japan Food Chemical Research Foundation, U.S. Food and Drug Administration (FDA), Ministry of Agriculture and Rural Affairs of China (GB 31,650–2019), India

strategies should exert emphasis on the minimization of antibiotic use to protect the aquaculture environments and to confirm the safety of consumers as well as farmers. Specific future thinking for the achievement of sustainable aquaculture production can be mentioned as follows:

– Development of effective vaccines, immunostimulants, or phage therapy alternatives to antibiotics to fight against diseases.
– Introduction of nutritious feed for the betterment of aqua-husbandry and maintaining of water quality in aquaculture industry.
– Proper education and training are essential for the sensible use of antibiotics to the aquaculture sectors worldwide.
– Implementation of Norwegian model, which requires professional's prescription to buy antibiotics in major aquaculture producers particularly in Asian and South American countries; however, it may not be feasible for large-scale aquaculture production countries like China and India. More insightful researches are needed to find out the suitable model to minimize the use of antibiotics in aquaculture of those countries.
– Improvement of farm support for the proper diagnosis and treatment of diseases.
– Making country-specific antibiotic usage and antimicrobial resistance data bank and insure it is accessible to all.
– Finally, emphasis on research, collaboration, harmonization of policies, regulations, and sharing information in regional and international bodies are extremely needed.

Conclusions

This review encompasses the present status of antibiotic use in aquaculture and their probable ecological, resistance, and human health risks and importantly current policy and regulatory frameworks and future concerns. Major aquaculture-producing countries such as China, Vietnam, India, Thailand, and Bangladesh are most likely using the elevated number of antibiotics in aquaculture. However, some of them have no stringent policy and regulatory frameworks or not strictly apply the regulations. Data related to antibiotic-resistant bacteria and resistance genes in aquaculture have also been accumulated in the current review. In addition, the maximum residue limits of antibiotics in fish/shellfish muscle in different countries gathered through many antibiotics are not yet to be decided. In addition, combined concentration of antibiotics in fish muscle is not considered in the current regulations worldwide. The main focus of this review is to develop a uniform regulation particularly for developing countries and to maintain the proper doses and number of antibiotic use in aquaculture for the production of safe, secure, and sustainable aqua-products. Finally, we suggest making a country-specific data bank of antibiotic usage, their doses, withdrawal period, and most importantly antibiotic resistance data bank for better management and to achieve sustainable aquaculture production in the future.

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